COMPUTER NETWORKS

1. Name two technologies by which you would connect two offices in remote locations.

Two technologies that would connect two offices in remote locations are **VPN** and **Cloud computing**.

2. What is internetworking?

Internetworking is a combination of two words, inter and networking which implies an association between totally different nodes or segments. This connection area unit is established through intercessor devices akin to routers or gateways. The first term for associate degree internetwork was interconnected. This interconnection is often among or between public, private, commercial, industrial, or governmental networks.
Thus, associate degree internetwork could be an assortment of individual networks, connected by intermediate networking devices, that function as one giant network. Internetworking refers to the trade, products, and procedures that meet the challenge of making and administering Internet works.

3. Name of the software layers or User support layer in the OSI model.

- **Application layer**
- **Presentation layer**
- **Session layer**

4. Name the hardware layers or network support layers in the OSI model.

- **Network layer**
- **Datalink layer**
- **Physical layer**

5. Define HTTPS protocol?

The full form of HTTPS is a Hypertext transfer protocol secure. It is an advanced version of the HTTP protocol. Its port number is 443 by default. It uses SSL/TLS protocol for providing security.

6. Name some services provided by the application layer in the Internet model?

Some services provided by the application layer in the Internet model are as follows:

- Mail services
- Directory services
- File transfer
- Access management
- Network virtual terminal

7. In which OSI layer is the header and trailer added?

At the Data link layer trailer is added and at the OSI model layer 6,5,4,3 added header.

8. What happens in the OSI model, as a data packet moves from the lower to upper layers?

In the OSI model, as a data packet moves from the lower to upper layers, headers get removed.

9. What happens in the OSI model, as a data packet moves from the upper to lower layers?

In the OSI model, as a data packet moves from the upper to lower layers, headers are added. This header contains useful information.

10. What is a zone-based firewall?

A Zone-based firewall is an advanced method of stateful firewall. In a stateful firewall, a stateful database is maintained in which the source IP address, destination IP address, source port number, and destination port number are recorded. Due to this, only the replies are allowed i.e. if the traffic is Generated

from inside the network then only the replies (of inside network traffic) coming from outside the network are allowed.

Cisco IOS router can be made firewall through two methods:

1. By using CBAC: create an access list and apply it to the interfaces keeping in mind what traffic should be allowed or denied and in what direction. This has an extra overhead for the administrator.
2. Using a Zone-based firewall.

11. What is a server farm?

A server farm is a set of many servers interconnected together and housed within the same physical facility. A server farm provides the combined computing power of many servers by simultaneously executing one or more applications or services. A server farm is generally a part of an enterprise data center or a component of a supercomputer. A server farm is also known as a server cluster or computer ranch.

12. Name the three means of user authentication.

There is biometrics (e.g. a thumbprint, iris scan), a token, or a password. There is also two-level authentication, which employs two of those methods.

13. What is Confidentiality, Integrity & Availability?

The **CIA triad** can be broadly defined as:
Confidentiality – means information is not disclosed to unauthorized individuals, entities, or processes. For example, if we say I have a password for my Gmail account but someone saw it while I was doing login into my Gmail account. In that case, my password has been compromised and Confidentiality has been breached.
Integrity – means maintaining the accuracy and completeness of data. This means data cannot be edited in an unauthorized way. For example, if an employee leaves an organization then in that case data for that employee in all departments like

accounts, should be updated to reflect the status to JOB LEFT so that data is complete and accurate in addition, this is only authorized persons should be allowed to edit employee data.
Availability – means information must be available when needed. For example, if one needs to access information about a particular employee to check whether an employee has outstood the number of leaves, that case, it requires collaboration from different organizational teams like network operations, development operations, incident response, and policy/change management.
Denial of service attack is one of the factors that can hamper the availability of information.

14. What is VPN?

VPN stands for the virtual private network. A virtual private network (VPN) is a technology that creates a safe and encrypted connection over a less secure network, such as the Internet. A Virtual Private Network is a way to extend a private network using a public network such as the Internet. The name only suggests that it is a Virtual “private network” i.e. user can be part of a local network sitting at a remote location. It makes use of tunneling protocols to establish a secure connection.

15. What is Symmetric and Asymmetric Encryption?

- **Symmetric Key Encryption: Encryption** is a process to change the form of any message in order to protect it from reading by anyone. In Symmetric-key encryption the message is encrypted by using a key and the same key is used to decrypt the message which makes it easy to use but less secure. It also requires a safe method to transfer the key from one party to another.
- **Asymmetric Key Encryption:** Asymmetric Key Encryption is based on public and private key encryption techniques. It uses two different keys to encrypt and decrypt the message. It is more secure than the symmetric key encryption technique but is much slower. For more details please refer **difference between symmetric and asymmetric encryption** articles.

16. What are Pvt. IP address?

The private IP address of a system is the **IP address** that is used to communicate within the same network. Using private IP data, information can be sent or received within the same network. For more details please refer **difference between private and public IP** addresses.

17. Explain does IP protects data on the network?

IP itself doesn't guarantee delivering data correctly. It leaves all data protection to the transport protocol. Both TCP and UDP have mechanisms that guarantee that the data they deliver to an application is correct.

Using the IP layer, the correct destination of the packet is identified & delivered. The Transport layer protocols(TCP/UDP) check if the data delivered is correct using the Checksum mechanism. However, if the destination IP is not alive, the packet is hopped by decreasing the TTL (Time to leave) field and when it becomes zero, the packet is lost and undelivered. If the transport layer is UDP, the source doesn't know of the failure in the delivery of the packet. For more details please refer **IP address**

18.Tell the name of the data Unit of the Transport layer?

The protocol data unit of the transport layer is a segment or datagram.

19. Tell the name of the data unit to send by the Internet layer?
The protocol data unit of the internet layer is a packet.

20.The difference between TCP and UDP protocols?

Transmission control protocol (TCP)	User datagram protocol (UDP)
TCP is a connection-oriented protocol. Connection-orientation means that the communicating devices should establish a connection before transmitting data and should close the connection after transmitting the data.	UDP is the datagram oriented protocol. This is because there is no overhead for opening a connection, maintaining a connection, and terminating a connection. UDP is efficient for broadcast and multicast types in order of network transmission.

Transmission control protocol (TCP)	User datagram protocol (UDP)
TCP is reliable as it guarantees the delivery of data to the destination router.	The delivery of data to the destination cannot be guaranteed in UDP.
TCP provides extensive error checking mechanisms. It is because it provides flow control and acknowledgement of data.	UDP has only the basic error checking mechanism using checksums.
Sequencing of data is a feature of Transmission Control Protocol (TCP). This means that packets arrive in-order at the receiver.	There is no sequencing of data in UDP. If the order is required, it has to be managed by the application layer.
TCP is comparatively slower than UDP.	UDP is faster, simpler, and more efficient than TCP.
Retransmission of lost packets is possible in TCP, but not in UDP.	There is no retransmission of lost packets in the User Datagram Protocol (UDP).
TCP has a (20-60) bytes variable length header.	UDP has an 8 bytes fixed-length header.
TCP is heavy-weight.	UDP is lightweight.
TCP doesn’t support Broadcasting.	UDP supports Broadcasting.
TCP is used by HTTP, HTTPs, FTP, SMTP and Telnet.	UDP is used by DNS, DHCP, TFTP, SNMP, RIP, and VoIP.

21.How TCP protocol provides reliability?

TCP is reliable as it uses checksum for error detection, attempts to recover lost or corrupted packets by re-transmission, acknowledgment policy, and timers. It uses features like byte numbers and sequence numbers and acknowledgment numbers so as to ensure reliability.

22.Write down the name of services provided by TCP?

- Process to process communication
- Stream orientation

- Full duplex service
- Multiplexing
- Reliability

23. What is the role of the TCP checksum field?

One of the important fields of TCP protocol format. It is 16 bits long. This field holds the checksum for error control. It is mandatory in TCP as opposed to UDP. For more details, please read **TCP/IP model** article.

24.What is PORT?

A port is basically a physical docking point that is basically used to connect the external devices to the computer, or we can say that a port acts as an interface between the computer and the external devices, e.g., we can connect hard drives and printers to the computer with the help of ports. For more details please refer to **various TCP and UDP ports** article.

25.Define the term Endpoint in TCP?

TcpEndpoint allows you to easily establish and communicate over TCP/IP network connections between client and server processes, possibly residing on different hosts. The TcpEndpoint class follows a telephone-like model of networking: clients “call” servers and servers “answer” clients. Once a network connection is established between a client and a server, the two can “talk” to each other by reading from and writing to the connection.

26.Explain the error control mechanism in TCP?

TCP protocol has methods for finding out corrupted segments, missing segments, out-of-order segments, and duplicated segments.
Error control in TCP is mainly done through the use of three simple techniques :

1.Checksum

2. Acknowledgment
3. Retransmission

27. What is round trip time?

The length of time taken by a data packet to be sent to a destination includes the time it takes for an acknowledgment of that packet to be received back at the original place.

28. What is the significance of TCP acknowledgments?

TCP acknowledgments are used to acknowledge packets that are successfully received by the host. The flag is set if the acknowledgment number field contains a valid acknowledgment number.

29. What is SCTP protocol?

SCTP stands for Stream Control Transmission Protocol. It is a connection-oriented protocol on computer networks that provides a full-duplex association, i.e., transmitting multiple streams of data between two endpoints at the same time that have established a connection in the network. It is sometimes referred to as next-generation TCP or TCPng. SCTP makes it easier to support telephonic conversation on the Internet. A telephonic conversation requires the transmitting of the voice along with other data at the same time on both ends. The SCTP protocol makes it easier to establish a reliable connection.SCTP is also intended to make it easier to establish connections over the wireless networks and managing the transmission of multimedia data. SCTP is a standard protocol (RFC 2960) and is developed by the Internet Engineering Task Force (IETF). For more details please read **the SCTP full form** article.

30. Explain the process of three-way handshaking protocol?

Process of three-way handshaking protocol

- Step 1 (SYN): In the first step, the client wants to establish a connection with the server, so it sends a segment with SYN(Synchronize Sequence Number) which informs the server that the client is likely to start communication and with what sequence number it starts segments.
- Step 2 (SYN + ACK): Server responds to the client request with the SYN-ACK signal bits set. Acknowledgment (ACK) signifies the response of the segment it received and SYN signifies with what sequence number it is likely to start the segment.
- Step 3 (ACK): In the final part, the client acknowledges the response of the server and they both establish a reliable connection with which they will start the actual data transfer.

31.What are the TCP connections phases?

In TCP, connection-oriented transmission requires three phases:

- Connection establishment
- Data Transfer
- Connection termination

32.Write features of TCP sliding window?

Features of the TCP sliding window:

- It uses variable-size windows to provide flow control as well as reliable transfer.
- TCP connections are full-duplex, two transfers proceed simultaneously over each connection, one in each direction.

- It support machine of various speed and size communicate through a network and also routers of various speeds.

33.What is the maximum size of the TCP header? What is the minimum size of TCP header?

- Maximum size of the TCP header = 60 bytes
- Minimum size of the TCP header = 20 bytes

34.Do port addresses need to be unique? Why or why not? Why are port addresses shorter than IP addresses?

Port addressing is done by the transport layer, which is the 4th layer of the OSI (Open System Interconnection) Model. Port addresses are short because they have to perform the end-to-end delivery of the message and the protocols are less in number than computer systems, therefore, port addresses are less than IP addresses. The IP address refers to the computer/device connected to the network and the port address refers to a particular protocol to communicate with the server as per client request. For more details read **TCP/IP ports and** its **application** article.

35.Are both UDP and IP unreliable to the same degree? Why or Why not?

No, UDP is unreliable and connectionless. While IP unreliability lies in the upper layer protocols. The IP packet becomes reliable if the upper layer protocol is TCP. In terms of data integrity, UDP is more reliable than IP. The checksum in the IP header only applies to the header itself, not to the whole packet. The UDP checksum applies to the whole user segment.

36.Define the term Datagram?

It is a logical unit to transfer information over the network. This data is transmitted from source to destination. It does not guarantee that data will be delivered or lost

on the way. It has two section headers and a data payload. It travels in-network without establishing a prior virtual network between source and destination., ie. unlike, it has no prior information about the path between source and destination. This data is frequently divided into smaller parts and transmitted to the defined route.

37.What are the Registered port and Dynamic port?

Registered port: The ports ranging from 1024 to 49151 are not assigned and controlled by an IANA. They can only be registered with IANA to prevent duplication.
Dynamic port: This port ranging from 49152 to 65535 is neither controlled nor registered. They can be used in any process.

38.Write the importance of TTL field?

The lifespan or lifetime of data that is being sent. Once, after that specified time is over or elapsed, the data will be discarded or it can also be stated as a number of hops that packet is set to exist in the network, after which that packet is discarded. The purpose of the TTL field is to avoid a situation in which an undeliverable datagram keeps circulating in the network.

39.An IPv4 packet has arrived with the first 8 bits as shown: 01000010 The receiver discards the packet. Why?

An IPv4 packet has arrived with the first 8 bits being 01000010 The receiver discards the packet because the first 4 bits represent version IPV4.
and another 4 bits represent header length (/ 4) which should range between 20 to 60 bytes. Here 0010 represents header length, is equal to 2 * 4 = 8. So, the receiver will reject the packet.

40.In an IPv4 packet, the value of HLEN is 1000 in binary. How many bytes of options are being carried by this packet?

If an IPv4 packet, the value of HLEN is 1000 in binary, then 32 bytes of options are carried by this packet.

HLEN = 1000 = 8

8*4 = 32

41.What are open loop congestion control and closed-loop congestion control techniques?

Open-loop congestion control policies are applied to prevent congestion before it happens. The congestion control is handled either by the source or the destination.

Closed-loop congestion control technique is used to treat or alleviate congestion after it happens.

42.Which fields of the IPv4 header change from router to router?
Total length field and header checksum of IPv4 header change from router to router.

43.The value of HLEN in an IPv4 datagram is 7.How many option bytes are present?

If the value of the HLEN field is 7, then there are 28 (since 7 × 4 = 28) bytes included in the header.

44.Can the value of the header length in an IPv4 packet be less than 5?

The value of the header length field of an IP packet can never be less than 5 because every IP datagram must have at least a base header that has a fixed size of 20 bytes.

45.A host is sending 100 datagrams to another host. If the identification number of the first datagram is 1024, what is the identification number of the last(in IPv4)?

if a source is sending 100 datagrams and for the first datagram identification no is 1024 then for the last datagram identification number will be 1024 + 99 = 1123.

46.Explain the reason for the elimination of the checksum in the IPv6 header?

The checksum is eliminated in IPv6 because it is provided by upper-layer protocols; it is therefore not needed at this level.

47.Write down the name of three strategies used to handle the transmission from IPv4 to IPv6?

- •Dual-stack
- •Tunneling
- •Header translation

48.What is tunneling?
A technique of internetworking called Tunneling is used when the source and destination networks of the same type are to be connected through a network of a different type.

49.An IP packet has arrived with the first 8 bits as 01000010. What is the version?

The first 4-bits show version i.e. 0100 which belongs to IP version 4.

50.What is fragmentation?

Fragmentation is an important function of the network layer. It is a technique in which gateways break up or divide larger packets into smaller ones called fragments. Each fragment is then sent as a separate internal packet. Each fragment has its own separate header and trailer. Sometimes, a fragmented datagram also gets fragmented when it encounters a network that handles smaller fragments. Thus, a datagram can be fragmented several times before it reaches its final destination. The reverse process of fragmentation is difficult. Reassembly of fragments is usually done by the destination host because each fragment has become an independent datagram.

OPERATING SYSTEM

1. What is a process and process table?

A process is an instance of a program in execution. For example, a Web Browser is a process, and a shell (or command prompt) is a process. The operating system is responsible for managing all the processes that are running on a computer and

allocates each process a certain amount of time to use the processor. In addition, the operating system also allocates various other resources that processes will need, such as computer memory or disks. To keep track of the state of all the processes, the operating system maintains a table known as the process table. Inside this table, every process is listed along with the resources the process is using and the current state of the process.

2. What are the different states of the process?

Processes can be in one of three states: running, ready, or waiting. The running state means that the process has all the resources it needs for execution and it has been given permission by the operating system to use the processor. Only one process can be in the running state at any given time. The remaining processes are either in a waiting state (i.e., waiting for some external event to occur such as user input or disk access) or a ready state (i.e., waiting for permission to use the processor). In a real operating system, the waiting and ready states are implemented as queues that hold the processes in these states.

3. What is a Thread?

A thread is a single sequence stream within a process. Because threads have some of the properties of processes, they are sometimes called lightweight processes. Threads are a popular way to improve the application through parallelism. For example, in a browser, multiple tabs can be different threads. MS Word uses multiple threads, one thread to format the text, another thread to process inputs, etc.

4. What are the differences between process and thread?

Threads are lightweight processes that share the same address space including the code section, data section and operating system resources such as the open files and signals. However, each thread has its own program counter (PC), register set and stack space allowing them to the execute independently within the same process context. Unlike **processes**, threads are not fully independent entities and can communicate and synchronize more efficiently making them suitable for the concurrent and parallel execution in the multi-threaded environment.

5. What are the benefits of multithreaded programming?

It makes the system more responsive and enables resource sharing. It leads to the use of multiprocess architecture. It is more economical and preferred.

6. What is Thrashing?

Thrashing is a situation when the performance of a computer degrades or collapses. Thrashing occurs when a system spends more time processing page faults than executing transactions. While processing page faults is necessary in order to appreciate the benefits of virtual memory, thrashing has a negative effect on the system. As the page fault rate increases, more transactions need processing from the paging device. The queue at the paging device increases, resulting in increased service time for a page fault.

7. What is Buffer?

A buffer is a memory area that stores data being transferred between two devices or between a device and an application.

8. What is virtual memory?

Virtual memory creates an illusion that each user has one or more contiguous address spaces, each beginning at address zero. The sizes of such virtual address spaces are generally very high. The idea of virtual memory is to use disk space to extend the RAM. Running processes don’t need to care whether the memory is from RAM or disk. The illusion of such a large amount of memory is created by subdividing the virtual memory into smaller pieces, which can be loaded into physical memory whenever they are needed by a process.

9. Explain the main purpose of an operating system?

An operating system acts as an intermediary between the user of a computer and computer hardware. The purpose of an operating system is to provide an environment in which a user can execute programs conveniently and efficiently.
An operating system is a software that manages computer hardware. The hardware must provide appropriate mechanisms to ensure the correct operation of the computer system and to prevent user programs from interfering with the proper operation of the system.

10. What is demand paging?

The process of loading the page into memory on demand (whenever a page fault occurs) is known as demand paging.

11. What is a kernel?

A kernel is the central component of an operating system that manages the operations of computers and hardware. It basically manages operations of memory and CPU time. It is a core component of an operating system. Kernel acts as a bridge between applications and data processing performed at the hardware level using inter-process communication and system calls.

12. What are the different scheduling algorithms?

First-Come, First-Served (FCFS) Scheduling.
Shortest-Job-Next (SJN) Scheduling.
Priority Scheduling.
Shortest Remaining Time.
Round Robin(RR) Scheduling.
Multiple-Level Queues Scheduling.

13. Describe the objective of multi-programming.

Multi-programming increases CPU utilization by organizing jobs (code and data) so that the CPU always has one to execute. The main objective of multi-programming is to keep multiple jobs in the main memory. If one job gets occupied with IO, the CPU can be assigned to other jobs.

14. What is the time-sharing system?

Time-sharing is a logical extension of multiprogramming. The CPU performs many tasks by switches that are so frequent that the user can interact with each program while it is running. A time-shared operating system allows multiple users to share computers simultaneously.

15. What problem we face in computer system without OS?

- Poor resource management
- Lack of User Interface
- No File System

- No Networking
- Error handling is big issue
- etc.

16. Give some benefits of multithreaded programming?

A thread is also known as a lightweight process. The idea is to achieve parallelism by dividing a process into multiple threads. Threads within the same process run in shared memory space,

17. Briefly explain FCFS.

FCFS stands for First Come First served. In the FCFS scheduling algorithm, the job that arrived first in the ready queue is allocated to the CPU and then the job that came second and so on. FCFS is a non-preemptive scheduling algorithm as a process that holds the CPU until it either terminates or performs I/O. Thus, if a longer job has been assigned to the CPU then many shorter jobs after it will have to wait.

18. What is the RR scheduling algorithm?

A round-robin scheduling algorithm is used to schedule the process fairly for each job in a time slot or quantum and interrupting the job if it is not completed by then the job comes after the other job which is arrived in the quantum time makes these scheduling fairly.

- Round-robin is cyclic in nature, so starvation doesn’t occur
- Round-robin is a variant of first-come, first-served scheduling
- No priority or special importance is given to any process or task
- RR scheduling is also known as Time slicing scheduling

19. Enumerate the different RAID levels?

A redundant array of independent disks is a set of several physical disk drives that the operating system sees as a single logical unit. It played a significant role in narrowing the gap between increasingly fast processors and slow disk drives. RAID has different levels:

- Level-0
- Level-1
- Level-2
- Level-3
- Level-4
- Level-5
- Level-6

20. What is Banker’s algorithm?

The banker’s algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation for the predetermined maximum possible amounts of all resources, then makes an “s-state” check to test for possible activities, before deciding whether allocation should be allowed to continue.

21. State the main difference between logical and physical address space?

Parameter	LOGICAL ADDRESS	PHYSICAL ADDRESS
Basic	generated by the CPU.	location in a memory unit.
Address Space	Logical Address Space is a set of all logical addresses generated by the CPU in reference to a program.	Physical Address is a set of all physical addresses mapped to the corresponding logical addresses.
Visibility	Users can view the logical address of a program.	Users can never view the physical address of the program.
Generation	generated by the CPU.	Computed by MMU.
Access	The user can use the logical address to access the physical address.	The user can indirectly access physical addresses but not directly.

22. How does dynamic loading aid in better memory space utilization?

With dynamic loading, a routine is not loaded until it is called. This method is especially useful when large amounts of code are needed in order to handle infrequently occurring cases such as error routines.

23. What are overlays?

The concept of overlays is that whenever a process is running it will not use the complete program at the same time, it will use only some part of it. Then overlay concept says that whatever part you required, you load it and once the part is done, then you just unload it, which means just pull it back and get the new part you required and run it. Formally, “The process of transferring a block of program code or other data into internal memory, replacing what is already stored”.

24. What is fragmentation?

Processes are stored and removed from memory, which makes free memory space, which is too little to even consider utilizing by different processes. Suppose, that process is not ready to dispense to memory blocks since its little size and memory hinder consistently staying unused is called fragmentation. This kind of issue occurs during a dynamic memory allotment framework when free blocks are small, so it can’t satisfy any request.

25. What is the basic function of paging?

Paging is a method or technique which is used for non-contiguous memory allocation. It is a fixed-size partitioning theme (scheme). In paging, both main memory and secondary memory are divided into equal fixed-size partitions. The partitions of the secondary memory area unit and the main memory area unit are known as pages and frames respectively.
Paging is a memory management method accustomed fetch processes from the secondary memory into the main memory in the form of pages. in paging, each process is split into parts wherever the size of every part is the same as the page size. The size of the last half could also be but the page size. The pages of the process area unit hold on within the frames of main memory relying upon their accessibility

26. How does swapping result in better memory management?

Swapping is a simple memory/process management technique used by the operating system(os) to increase the utilization of the processor by moving some blocked processes from the main memory to the secondary memory thus forming a queue of the temporarily suspended processes and the execution continues with the newly arrived process. During regular intervals that are set by the operating system, processes can be copied from the main memory to a backing store and then copied back later. Swapping allows more processes to be run that can fit into memory at one time

27. Write a name of classic synchronization problems?
    - Bounded-buffer
    - Readers-writers
    - Dining philosophers
    - Sleeping barber
28. What is the Direct Access Method?

The direct Access method is based on a disk model of a file, such that it is viewed as a numbered sequence of blocks or records. It allows arbitrary blocks to be read or written. Direct access is advantageous when accessing large amounts of information. Direct memory access (DMA) is a method that allows an input/output (I/O) device to send or receive data directly to or from the main memory, bypassing the CPU to speed up memory operations. The process is managed by a chip known as a DMA controller (DMAC).

29. When does thrashing occur?

Thrashing occurs when processes on the system frequently access pages, not available memory.

30. What is the best page size when designing an operating system?

The best paging size varies from system to system, so there is no single best when it comes to page size. There are different factors to consider in order to come up

with a suitable page size, such as page table, paging time, and its effect on the overall efficiency of the operating system.

31. What is multitasking?

Multitasking is a logical extension of a multiprogramming system that supports multiple programs to run concurrently. In multitasking, more than one task is executed at the same time. In this technique, the multiple tasks, also known as processes, share common processing resources such as a CPU.

32. What is caching?

The cache is a smaller and faster memory that stores copies of the data from frequently used main memory locations. There are various different independent caches in a CPU, which store instructions and data. Cache memory is used to reduce the average time to access data from the Main memory.

33. What is spooling?

Spooling refers to simultaneous peripheral operations online, spooling refers to putting jobs in a buffer, a special area in memory, or on a disk where a device can access them when it is ready. Spooling is useful because devices access data at different rates.

34. What is the functionality of an Assembler?

The Assembler is used to translate the program written in Assembly language into machine code. The source program is an input of an assembler that contains assembly language instructions. The output generated by the assembler is the object code or machine code understandable by the computer.

35. What are interrupts?

The interrupts are a signal emitted by hardware or software when a process or an event needs immediate attention. It alerts the processor to a high-priority process requiring interruption of the current working process. In I/O devices one of the bus control lines is dedicated to this purpose and is called the Interrupt Service Routine (ISR).

36. What is GUI?

GUI is short for Graphical User Interface. It provides users with an interface wherein actions can be performed by interacting with icons and graphical symbols.

37. What is preemptive multitasking?

Preemptive multitasking is a type of multitasking that allows computer programs to share operating systems (OS) and underlying hardware resources. It divides the overall operating and computing time between processes, and the switching of resources between different processes occurs through predefined criteria.

38. What is a pipe and when is it used?

A Pipe is a technique used for inter-process communication. A pipe is a mechanism by which the output of one process is directed into the input of another process. Thus it provides a one-way flow of data between two related processes.

39. What are the advantages of semaphores?

- They are machine-independent.
- Easy to implement.
- Correctness is easy to determine.
- Can have many different critical sections with different semaphores.
- Semaphores acquire many resources simultaneously.
- No waste of resources due to busy waiting.

40. What is a bootstrap program in the OS?

Bootstrapping is the process of loading a set of instructions when a computer is first turned on or booted. During the startup process, diagnostic tests are performed, such as the power-on self-test (POST), which set or checks configurations for devices and implements routine testing for the connection of peripherals, hardware, and external memory devices. The bootloader or bootstrap program is then loaded to initialize the OS.

41. What is IPC?

Inter-process communication (IPC) is a mechanism that allows processes to communicate with each other and synchronize their actions. The communication between these processes can be seen as a method of cooperation between them.

42. What are the different IPC mechanisms?

These are the methods in IPC:

- **Pipes (Same Process)**: This allows a flow of data in one direction only. Analogous to simplex systems (Keyboard). Data from the output is usually buffered until the input process receives it which must have a common origin.
- **Named Pipes (Different Processes)**: This is a pipe with a specific name it can be used in processes that don't have a shared common process origin. E.g. FIFO where the details written to a pipe are first named.
- **Message Queuing**: This allows messages to be passed between processes using either a single queue or several message queues. This is managed by the system kernel these messages are coordinated using an API.
- **Semaphores**: This is used in solving problems associated with synchronization and avoiding race conditions. These are integer values that are greater than or equal to 0.
- **Shared Memory**: This allows the interchange of data through a defined area of memory. Semaphore values have to be obtained before data can get access to shared memory.
- **Sockets**: This method is mostly used to communicate over a network between a client and a server. It allows for a standard connection which is computer and OS independent

43. What is the difference between preemptive and non-preemptive scheduling?

- In preemptive scheduling, the CPU is allocated to the processes for a limited time whereas, in Non-preemptive scheduling, the CPU is allocated to the process till it terminates or switches to waiting for the state.

•The executing process in preemptive scheduling is interrupted in the middle of execution when a higher priority one comes whereas, the executing process in non-preemptive scheduling is not interrupted in the middle of execution and waits till its execution.

•In Preemptive Scheduling, there is the overhead of switching the process from the ready state to the running state, vice-verse, and maintaining the ready queue. Whereas the case of non-preemptive scheduling has no overhead of switching the process from running state to ready state.

•In preemptive scheduling, if a high-priority process frequently arrives in the ready queue then the process with low priority has to wait for a long, and it may have to starve. On the other hand, in non-preemptive scheduling, if CPU is allocated to the process having a larger burst time then the processes with a small burst time may have to starve.

•Preemptive scheduling attains flexibility by allowing the critical processes to access the CPU as they arrive in the ready queue, no matter what process is executing currently. Non-preemptive scheduling is called rigid as even if a critical process enters the ready queue the process running CPU is not disturbed.

•Preemptive Scheduling has to maintain the integrity of shared data that’s why it is cost associative which is not the case with Non-preemptive Scheduling.

44. What is the zombie process?

A process that has finished the execution but still has an entry in the process table to report to its parent process is known as a zombie process. A child process always first becomes a zombie before being removed from the process table. The parent process reads the exit status of the child process which reaps off the child process entry from the process table.

45. What are orphan processes?

A process whose parent process no more exists i.e. either finished or terminated without waiting for its child process to terminate is called an orphan process.
46. What are starvation and aging in OS?

Starvation: Starvation is a resource management problem where a process does not get the resources it needs for a long time because the resources are being allocated to other processes.
Aging: Aging is a technique to avoid starvation in a scheduling system. It works by adding an aging factor to the priority of each request. The aging factor must increase the priority of the request as time passes and must ensure that a request will eventually be the highest priority request
47. Write about monolithic kernel?

Apart from microkernel, Monolithic Kernel is another classification of Kernel. Like microkernel, this one also manages system resources between application and hardware, but user services and kernel services are implemented under the same address space. It increases the size of the kernel, thus increasing the size of an operating system as well. This kernel provides CPU scheduling, memory management, file management, and other operating system functions through system calls. As both services are implemented under the same address space, this makes operating system execution faster.
48. What is Context Switching?

Switching of CPU to another process means saving the state of the old process and loading the saved state for the new process. In Context Switching the process is stored in the Process Control Block to serve the new process so that the old process can be resumed from the same part it was left.
49. What is the difference between the Operating system and kernel?

Operating System	Kernel
Operating System is system software.	The kernel is system software that is part of the Microkerneloperating system.
Operating System provides an interface b/w the user and the hardware.	The kernel provides an interface b/w the application and hardware.
It also provides protection and security.	Its main purpose is memory management, disk

Operating System	Kernel
	management, process management and task management.
All systems need a real-time operating system and a microkernel system to run.	All operating system needs a kernel to run.
Type of operating system includes single and multiuser OS, multiprocessor OS, real-time OS, Distributed OS.	Type of kernel includes Monolithic and Microkernel.
It is the first program to load when the computer boots up.	It is the first program to load when the operating system loads

50. What is the difference between process and thread?

S.NO	Process	Thread
1.	Process means any program is in execution.	Thread means a segment of a process.
2.	The process is less efficient in terms of communication.	Thread is more efficient in terms of communication.
3.	The process is isolated.	Threads share memory.
4.	The process is called heavyweight the process.	Thread is called lightweight process.
5.	Process switching uses, another process interface in operating system.	Thread switching does not require to call an operating system and cause an interrupt to the kernel.
6.	If one process is blocked then it will not affect the execution of other process	The second, thread in the same task could not run, while one server thread is blocked.
7.	The process has its own Process Control Block, Stack and Address Space.	Thread has Parents' PCB, its own Thread Control Block and Stack and common Address space.

51. What is PCB?

the process control block (PCB) is a block that is used to track the process's execution status. A process control block (PCB) contains information about the process, i.e. registers, quantum, priority, etc. The process table is an array of PCBs, that means logically contains a PCB for all of the current processes in the system.

52. When is a system in a safe state?

The set of dispatchable processes is in a safe state if there exists at least one temporal order in which all processes can be run to completion without resulting in a deadlock.

53. What is Cycle Stealing?

cycle stealing is a method of accessing computer memory (RAM) or bus without interfering with the CPU. It is similar to direct memory access (DMA) for allowing I/O controllers to read or write RAM without CPU intervention.

54. What are a Trap and Trapdoor?

A trap is a software interrupt, usually the result of an error condition, and is also a non-maskable interrupt and has the highest priority Trapdoor is a secret undocumented entry point into a program used to grant access without normal methods of access authentication.

55. Write a difference between process and program?

S.NO	Program	Process
1.	Program contains a set of instructions designed to complete a specific task.	Process is an instance of an executing program.
2.	Program is a passive entity as it resides in the secondary memory.	Process is anThe process active entity as it is created during execution and loaded into the main memory.

S.NO	Program	Process
3.	The program exists in a single place and continues to exist until it is deleted.	Process exists for a limited span of time as it gets terminated after the completion of a task.
4.	A program is a static entity.	The process is a dynamic entity.
5.	Program does not have any resource requirement, it only requires memory space for storing the instructions.	Process has a high resource requirement, it needs resources like CPU, memory address, and I/O during its lifetime.
6.	The program does not have any control block.	The process has its own control block called Process Control Block.

56. What is a dispatcher?

The dispatcher is the module that gives process control over the CPU after it has been selected by the short-term scheduler. This function involves the following:

- Switching context
- Switching to user mode
- Jumping to the proper location in the user program to restart that program

57. Define the term dispatch latency?

Dispatch latency can be described as the amount of time it takes for a system to respond to a request for a process to begin operation. With a scheduler written specifically to honor application priorities, real-time applications can be developed with a bounded dispatch latency.

58. What are the goals of CPU scheduling?

- Max CPU utilization [Keep CPU as busy as possible]Fair allocation of CPU.
- Max throughput [Number of processes that complete their execution per time unit]
- Min turnaround time [Time taken by a process to finish execution]
- Min waiting time [Time a process waits in ready queue]
- Min response time [Time when a process produces the first response]

59. What is a critical- section?

When more than one processes access the same code segment that segment is known as the critical section. The critical section contains shared variables or resources which are needed to be synchronized to maintain the consistency of data variables. In simple terms, a critical section is a group of instructions/statements or regions of code that need to be executed atomically such as accessing a resource (file, input or output port, global data, etc.).
60. Write the name of synchronization techniques?

- •Mutexes
- •Condition variables
- •Semaphores
- •File locks

Intermediate OS Interview Questions

61. Write a difference between a user-level thread and a kernel-level thread?

User-level thread	Kernel level thread
User threads are implemented by users.	kernel threads are implemented by OS.
OS doesn’t recognize user-level threads.	Kernel threads are recognized by OS.
Implementation of User threads is easy.	Implementation of the perform kernel thread is complicated.
Context switch time is less.	Context switch time is more.
Context switch requires no hardware support.	Hardware support is needed.
If one user-level thread performs a blocking operation then entire process will be blocked.	If one kernel thread perform a the blocking operation then another thread can continue execution.
User-level threads are designed as dependent	Kernel level threads are designed as

User-level thread	Kernel level thread
threads.	independent threads.

62. Write down the advantages of multithreading?

Some of the most important benefits of MT are:

•Improved throughput. Many concurrent compute operations and I/O requests within a single process.

•Simultaneous and fully symmetric use of multiple processors for computation and I/O.

•Superior application responsiveness. If a request can be launched on its own thread, applications do not freeze or show the “hourglass”. An entire application will not block or otherwise wait, pending the completion of another request.

•Improved server responsiveness. Large or complex requests or slow clients don’t block other requests for service. The overall throughput of the server is much greater.

•Minimized system resource usage. Threads impose minimal impact on system resources. Threads require less overhead to create, maintain, and manage than a traditional process.

•Program structure simplification. Threads can be used to simplify the structure of complex applications, such as server-class and multimedia applications. Simple routines can be written for each activity, making complex programs easier to design and code, and more adaptive to a wide variation in user demands.

•Better communication. Thread synchronization functions can be used to provide enhanced process-to-process communication. In addition, sharing large amounts of data through separate threads of execution within the same address space provides extremely high-bandwidth, low-latency communication between separate tasks within an application

63. Difference between Multithreading and Multitasking?

S.No	Multi-threading	Multi-tasking
1.	Multiple threads are executing at the same time at the same or different part of the program.	Several programs are executed concurrently.
2.	CPU switches between multiple threads.	CPU switches between multiple tasks and processes.
3.	It is the process of a lightweight part.	It is a heavyweight process.
4.	It is a feature of the process.	It is a feature of the OS.
5.	Multi-threading is sharing of computing resources among threads of a single process.	Multitasking is sharing of computing resources(CPU, memory, devices, etc.) among processes.

64. What are the drawbacks of semaphores?

- Priority Inversion is a big limitation of semaphores.
- Their use is not enforced but is by convention only.
- The programmer has to keep track of all calls to wait and signal the semaphore.
- With improper use, a process may block indefinitely. Such a situation is called Deadlock.

65. What is Peterson's approach?

It is a concurrent programming algorithm. It is used to synchronize two processes that maintain the mutual exclusion for the shared resource. It uses two variables, a bool array flag of size 2 and an int variable turn to accomplish it.

66. Define the term Bounded waiting?

A system is said to follow bounded waiting conditions if a process wants to enter into a critical section will enter in some finite time.

67. What are the solutions to the critical section problem?

There are three solutions to the critical section problem:

- Software solutions

- Hardware solutions
- Semaphores

68. What is a Banker’s algorithm?

The banker’s algorithm is a resource allocation and deadlock avoidance algorithm that tests for safety by simulating the allocation for the predetermined maximum possible amounts of all resources, then makes an “s-state” check to test for possible activities, before deciding whether allocation should be allowed to continue.

69. What is concurrency?

A state in which a process exists simultaneously with another process than those it is said to be concurrent.

70. Write a drawback of concurrency?

- It is required to protect multiple applications from one another.
- It is required to coordinate multiple applications through additional mechanisms.
- Additional performance overheads and complexities in operating systems are required for switching among applications.
- Sometimes running too many applications concurrently leads to severely degraded performance.

71. What are the necessary conditions which can lead to a deadlock in a system?

Mutual Exclusion: There is a resource that cannot be shared.
Hold and Wait: A process is holding at least one resource and waiting for another resource, which is with some other process.
No Preemption: The operating system is not allowed to take a resource back from a process until the process gives it back.
Circular Wait: A set of processes waiting for each other in circular form.

72. What are the issues related to concurrency?

- **Non-atomic**: Operations that are non-atomic but interruptible by multiple processes can cause problems.
- **Race conditions**: A race condition occurs of the outcome depends on which of several processes gets to a point first.
- **Blocking**: Processes can block waiting for resources. A process could be blocked for a long period of time waiting for input from a terminal. If the process is required to periodically update some data, this would be very undesirable.
- **Starvation**: It occurs when a process does not obtain service to progress.
- **Deadlock**: It occurs when two processes are blocked and hence neither can proceed to execute

73. Why do we use precedence graphs?

A precedence graph is a directed acyclic graph that is used to show the execution level of several processes in the operating system. It has the following properties also:

- Nodes of graphs correspond to individual statements of program code.
- An edge between two nodes represents the execution order.
- A directed edge from node A to node B shows that statement A executes first and then Statement B executes

74. Explain the resource allocation graph?

The resource allocation graph is explained to us what is the state of the system in terms of processes and resources. One of the advantages of having a diagram is, sometimes it is possible to see a deadlock directly by using RAG.

75. What is a deadlock?

Deadlock is a situation when two or more processes wait for each other to finish and none of them ever finish. Consider an example when two trains are coming toward each other on the same track and there is only one track, none of the trains can move once they are in front of each other. A similar situation occurs in

operating systems when there are two or more processes that hold some resources and wait for resources held by other(s).

76. What is the goal and functionality of memory management?

The goal and functionality of memory management are as follows;

- •Relocation
- •Protection
- •Sharing
- •Logical organization
- •Physical organization

77. Write a difference between physical address and logical address?

S.NO.	Parameters	Logical address	Physical Address
1.	Basic	It is the virtual address generated by CPU.	The physical address is a location in a memory unit.
2.	Address	Set of all logical addresses generated by the CPU in reference to a program is referred to as Logical Address Space.	Set of all physical addresses mapped to the corresponding logical addresses is referred to as a Physical Address.
3.	Visibility	The user can view the logical address of a program.	The user can never view the physical address of the program
4.	Access	The user uses the logical address to access the physical address.	The user can not directly access the physical address
5.	Generation	The Logical Address is generated by the CPU	Physical Address is Computed by MMU

78. Explain address binding?

The Association of program instruction and data to the actual physical memory locations is called Address Binding.

79. Write different types of address binding?

Address Binding is divided into three types as follows.

•Compile-time Address Binding

•**Load time Address Binding**

•**Execution time Address Binding**

80. Write an advantage of dynamic allocation algorithms?

•When we do not know how much amount of memory would be needed for the program beforehand.

•When we want data structures without any upper limit of memory space.

•When you want to use your memory space more efficiently.

•Dynamically created lists insertions and deletions can be done very easily just by the manipulation of addresses whereas in the case of statically allocated memory insertions and deletions lead to more movements and wastage of memory.

•When you want to use the concept of structures and linked lists in programming, dynamic memory allocation is a must

81. Write a difference between internal fragmentation and external fragmentation?

S.NO	Internal fragmentation	External fragmentation
1.	In internal fragmentation fixed-sized memory, blocks square measure appointed to process.	In external fragmentation, variable-sized memory blocks square measure appointed to the method.
2.	Internal fragmentation happens when the method or process is larger than the memory.	External fragmentation happens when the method or process is removed.
3.	The solution to internal fragmentation is the best-fit block.	Solution for external fragmentation is compaction, paging and segmentation.
4.	Internal fragmentation occurs when memory is divided into fixed-sized partitions.	External fragmentation occurs when memory is divided into variable-size partitions based on the size of processes.
5.	The difference between memory	The unused spaces formed between non-

S.NO	Internal fragmentation	External fragmentation
	allocated and required space or memory is called Internal fragmentation.	contiguous memory fragments are too small to serve a new process, which is called External fragmentation.

82. Define the Compaction?

The process of collecting fragments of available memory space into contiguous blocks by moving programs and data in a computer's memory or disk.

83. Write about the advantages and disadvantages of a hashed-page table?

Advantages

- The main advantage is synchronization.
- In many situations, hash tables turn out to be more efficient than search trees or any other table lookup structure. For this reason, they are widely used in many kinds of computer software, particularly for associative arrays, database indexing, caches, and sets.

Disadvantages

- Hash collisions are practically unavoidable. when hashing a random subset of a large set of possible keys.
- Hash tables become quite inefficient when there are many collisions.
- Hash table does not allow null values, like a hash map.
- Define Compaction.

84. Write a difference between paging and segmentation?

S.NO	Paging	Segmentation
1.	In paging, program is divided into	In segmentation, the program is divided into

S.NO	Paging	Segmentation
	fixed or mounted-size pages.	variable-size sections.
2.	For the paging operating system is accountable.	For segmentation compiler is accountable.
3.	Page size is determined by hardware.	Here, the section size is given by the user.
4.	It is faster in comparison of segmentation.	Segmentation is slow.
5.	Paging could result in internal fragmentation.	Segmentation could result in external fragmentation.
6.	In paging, logical address is split into that page number and page offset.	Here, logical address is split into section number and section offset.
7.	Paging comprises a page table which encloses the base address of every page.	While segmentation also comprises the segment table which encloses the segment number and segment offset.
8.	A page table is employed to keep up the page data.	Section Table maintains the section data.
9.	In paging, operating system must maintain a free frame list.	In segmentation, the operating system maintains a list of holes in the main memory.
10.	Paging is invisible to the user.	Segmentation is visible to the user.
11.	In paging, processor needs page number, offset to calculate the absolute address.	In segmentation, the processor uses segment number, and offset to calculate the full address.

85. Write a definition of Associative Memory and Cache Memory?

S.No.	Associative Memory	Cache Memory
1	A memory unit accessed by content is	Fast and small memory is called cache

S.No.	Associative Memory	Cache Memory
	called associative memory.	memory.
2	It reduces the time required to find the item stored in memory.	It reduces the average memory access time.
3	Here data is accessed by its content.	Here, data are accessed by their address.
4	It is used where search time is very short.	It is used when a particular group of data is accessed repeatedly.
5	Its basic characteristic is its logic circuit for matching its content.	Its basic characteristic is its fast access

86. What is “Locality of reference”?

The locality of reference refers to a phenomenon in which a computer program tends to access the same set of memory locations for a particular time period. In other words, Locality of Reference refers to the tendency of the computer program to access instructions whose addresses are near one another.

87. Write down the advantages of virtual memory?

- A higher degree of multiprogramming.
- Allocating memory is easy and cheap
- Eliminates external fragmentation
- Data (page frames) can be scattered all over the PM
- Pages are mapped appropriately anyway
- Large programs can be written, as the virtual space available is huge compared to physical memory.
- Less I/O required leads to faster and easy swapping of processes.
- More physical memory is available, as programs are stored on virtual memory, so they occupy very less space on actual physical memory.
- More efficient swapping

88. How to calculate performance in virtual memory?

The performance of virtual memory of a virtual memory management system depends on the total number of page faults, which depend on “paging policies” and “frame allocation”.
Effective access time = (1-p) x Memory access time + p x page fault time
89. Write down the basic concept of the file system?

A file is a collection of related information that is recorded on secondary storage. Or file is a collection of logically related entities. From the user’s perspective, a file is the smallest allotment of logical secondary storage.
90. Write the names of different operations on file?

Operation on file:

- Create
- Open
- Read
- Write
- Rename
- Delete
- Append
- Truncate
- Close

91. Define the term Bit-Vector?

A Bitmap or Bit Vector is a series or collection of bits where each bit corresponds to a disk block. The bit can take two values: 0 and 1: 0 indicates that the block is allocated and 1 indicates a free block.
92. What is a File allocation table?

FAT stands for File Allocation Table and this is called so because it allocates different files and folders using tables. This was originally designed to handle small file systems and disks. A file allocation table (FAT) is a table that an operating system maintains on a hard disk that provides a map of the cluster (the basic units of logical storage on a hard disk) that a file has been stored in.

93. What is rotational latency?

Rotational Latency: Rotational Latency is the time taken by the desired sector of the disgeek to rotate into a position so that it can access the read/write heads. So the disk scheduling algorithm that gives minimum rotational latency is better.
94. What is seek time?

Seek Time: Seek time is the time taken to locate the disk arm to a specified track where the data is to be read or written. So the disk scheduling algorithm that gives a minimum average seek time is better.

DATABASE MANAGEMENT SYSTEM

1. What are the advantages of DBMS over traditional file-based systems?

Database management systems were developed to handle the following difficulties of typical File-processing systems supported by conventional operating systems.
1. Data redundancy and inconsistency
2. Difficulty in accessing data
3. Data isolation – multiple files and formats
4. Integrity problems
5. Atomicity of updates
6. Concurrent access by multiple users
7. Security problems

2. What are super, primary, candidate, and foreign keys?

A **super key** is a set of attributes of a relation schema upon which all attributes of the schema are functionally dependent. No two rows can have the same value of super key attributes.
A **Candidate key** is a minimal superkey, i.e., no proper subset of Candidate key attributes can be a superkey.
A **Primary Key** is one of the candidate keys. One of the candidate keys is selected

as most important and becomes the primary key. There cannot be more than one primary key in a table..
A Foreign key is a field (or collection of fields) in one table that uniquely identifies a row of another table.
3. What is the difference between primary key and unique constraints?

The primary key cannot have NULL value, the unique constraints can have NULL values. There is only one primary key in a table, but there can be multiple unique constrains.

4.What is database normalization?

It is a process of analyzing the given relation schemas based on their functional dependencies and primary keys to achieve the following desirable properties:
1. Minimizing Redundancy
2. Minimizing the Insertion, Deletion, And Update Anomalies Relation schemas that do not meet the properties are decomposed into smaller relation schemas that could meet desirable properties.

5. Why is the use of DBMS recommended? Explain by listing some of its major advantages?

Some of the major advantages of DBMS are as follows:

- Controlled Redundancy: DBMS supports a mechanism to control the redundancy of data inside the database by integrating all the data into a single database and as data is stored in only one place, the duplicity of data does not happen.
- Data Sharing: Sharing of data among multiple users simultaneously can also be done in DBMS as the same database will be shared among all the users and by different application programs.
- Backup and Recovery Facility: DBMS minimizes the pain of creating the backup of data again and again by providing a feature of ‘backup and recovery’

which automatically creates the data backup and restores the data whenever required.

•Enforcement of Integrity Constraints: Integrity Constraints are very important to be enforced on the data so that the refined data after putting some constraints are stored in the database and this is followed by DBMS.

•Independence of Data: It simply means that you can change the structure of the data without affecting the structure of any of the application programs.

6. What are the differences between DDL, DML, and DCL in SQL?

Following are some details of three :
DDL stands for Data Definition Language. SQL queries like CREATE, ALTER, DROP, TRUNCATE and RENAME come under this.
DML stands for Data Manipulation Language. SQL queries like SELECT, INSERT, DELETE and UPDATE come under this.
DCL stands for Data Control Language. SQL queries like GRANT and REVOKE come under this.

7. What is the difference between having and where clause?

HAVING is used to specify a condition for a group or an aggregate function used in a select statement. The WHERE clause selects before grouping. The HAVING clause selects rows after grouping. Unlike the HAVING clause, the WHERE clause cannot contain aggregate functions. (See **this** for examples). See **Having vs Where Clause?** for more details
8.How to print duplicate rows in a table?

See **https://www.geeksforgeeks.org/how-to-print-duplicate-rows-in-a-table/**
9. What is Join?

An SQL Join is used to combine data from two or more tables, based on a common field between them. For example, consider the following two tables.

10.What is a view in SQL? How to create a view?

A view is a virtual table based on the result-set of an SQL statement. We can create it using create view syntax.

11.What are the uses of view?

1. Views can represent a subset of the data contained in a table; consequently, a view can limit the degree of exposure of the underlying tables to the outer world: a given user may have permission to query the view, while denied access to the rest of the base table.
2. Views can join and simplify multiple tables into a single virtual table.
3. Views can act as aggregated tables, where the database engine aggregates data (sum, average, etc.) and presents the calculated results as part of the data.
4. Views can hide the complexity of data.
5. Views take very little space to store; the database contains only the definition of a view, not a copy of all the data which it presents.
6. Depending on the SQL engine used, views can provide extra security.
12. What is a Trigger?

A Trigger is a code associated with insert, update or delete operations. The code is executed automatically whenever the associated query is executed on a table. Triggers can be useful to maintain integrity in the database.
13. What is a stored procedure?

A stored procedure is like a function that contains a set of operations compiled together. It contains a set of operations that are commonly used in an application to do some common database tasks.
14. What is the difference between Trigger and Stored Procedure?

Unlike Stored Procedures, Triggers cannot be called directly. They can only be associated with queries.
15. What is a transaction? What are ACID properties?

A Database Transaction is a set of database operations that must be treated as a whole, which means either all operations are executed or none of them. An example can be a bank transaction from one account to another account. Either both debit and credit operations must be executed or none of

them. ACID (Atomicity, Consistency, Isolation, Durability) is a set of properties that guarantee that database transactions are processed reliably.

16. What are indexes?

A database index is a data structure that improves the speed of data retrieval operations on a database table at the cost of additional writes and the use of more storage space to maintain the extra copy of data. Data can be stored only in one order on a disk. To support faster access according to different values, faster search like binary search for different values is desired, For this purpose, indexes are created on tables. These indexes need extra space on the disk, but they allow faster search according to different frequently searched values.

17. What are clustered and non-clustered Indexes?

Clustered indexes are the index according to which data is physically stored on a disk. Therefore, only one clustered index can be created on a given database table.

Non-clustered indexes don’t define the physical ordering of data, but logical ordering. Typically, a tree is created whose leaf point to disk records. B-Tree or B+ tree are used for this purpose.

18. What is Denormalization?

Denormalization is a database optimization technique in which we add redundant data to one or more tables.

19. What is CLAUSE in SQL?

A clause in SQL is a part of a query that lets you filter or customize how you want your data to be queried to you.

20. What is a Live Lock?

Livelock situation can be defined as when two or more processes continually repeat the same interaction in response to changes in the other processes without doing any useful work These processes are not in the waiting state, and they are running concurrently. This is different from a deadlock because in a deadlock all processes are in the waiting state.

21. What is QBE?

Query-by-example represents a visual/graphical approach for accessing information in a database through the use of query templates called skeleton tables. It is used by entering example values directly into a query template to represent what is to be achieved. QBE is used by many database systems for personal computers. QBE is a very powerful facility that gives the user the capability to access the information a user wants without the knowledge of any programming language. Queries in QBE are expressed by skeleton tables.

22. Why are cursors necessary in embedded SQL?

A cursor is an object used to store the output of a query for row-by-row processing by the application programs. SQL statements operate on a set of data and return a set of data. On other hand, host language programs operate on a row at a time. The cursors are used to navigate through a set of rows returned by an embedded SQL SELECT statement. A cursor can be compared to a pointer.

23. What is the purpose of normalization in DBMS?

Database normalization is the process of organizing the attributes of the database to reduce or eliminate data redundancy (having the same data but at different places).

24. What is the difference between a database schema and a database state?

The collection of information stored in a database at a particular moment in time is called database state while the overall design of the database is called the database schema.

25. What is the purpose of SQL?

SQL stands for Structured Query Language whose main purpose is to interact with the relational databases in the form of inserting, deleting and updating/modifying the data in the database.

26. Explain the concepts of a Primary key and Foreign Key.
Primary Key is used to uniquely identify the records in a database table
while Foreign Key is mainly used to link two or more tables together, as this is a particular field(s) in one of the database tables which are the primary key of some other table.

Example: There are 2 tables – Employee and Department. Both have one common field/column as ‘ID’ where ID is the primary key of the Employee table while this is the foreign key for the Department table.

27.What are the main differences between Primary key and Unique Key?
Given below are few differences:

The main difference between the Primary key and the Unique key is that the Primary key can never have a null value while the Unique key may consist of a null value.

28. What is the concept of sub-query in terms of SQL?

Sub-query is basically the query that is included inside some other query and can also be called an inner query which is found inside the outer query.

29. What is the use of the DROP command and what are the differences between DROP, TRUNCATE and DELETE commands?

DROP command is a DDL command which is used to drop/delete the existing table, database, index, or view from the database.

The major difference between DROP, TRUNCATE and DELETE commands are:

DROP and TRUNCATE commands are the DDL commands which are used to delete tables from the database.

And when we make use of a DROP command, the tables get deleted permanently all the privileges and indexes that are related to the table also get deleted. This operation cannot be rolled back and so should be used only when necessary.

However in case of TRUNCATE, only the data stored in a table is deleted and the structure of the table is preserved and you can re-insert data by the use of “*INSERT INTO clause”.* It can be rolled back until the commit has been made. DELETE command, on the other hand, is a DML Command which is used to delete rows from the table and this can be rolled back, however its considered slower than truncate. Using the delete command, we can delete 1 or more specific rows from the table.

30. What is the main difference between UNION and UNION ALL?

UNION and UNION ALL are used to join the data from 2 or more tables but UNION removes duplicate rows and picks the rows which are distinct after combining the data from the tables whereas UNION ALL does not remove the duplicate rows, it just picks all the data from the tables.

31. What is Correlated Subquery in DBMS?

A Subquery is also known as a nested query i.e. a query written inside some query. When a Subquery is executed for each of the rows of the outer query then it is termed as a Correlated Subquery.

32. Explain Entity, Entity Type, and Entity Set in DBMS?

The entity is an object, place, or thing which has its independent existence in the real world and about which data can be stored in a database. For Example, any person, book, etc.

Entity Type is a collection of entities that have the same attributes. For Example, the STUDENT table contains rows in which each row is an entity holding the attributes like name, age, and id of the students, hence STUDENT is an Entity Type that holds the entities having the same attributes.

33. What are the different levels of abstraction in the DBMS?

There are 3 levels of data abstraction in the DBMS.

They include:

Physical Level: This is the lowest level of the data abstraction which states how the data is stored in the database.

Logical Level: This is the next level of the data abstraction which states the type of the data and the relationship among the data that is stored in the database.

View Level: This is the highest level in the data abstraction which shows/states only a part of the database.

34 . What integrity rules exist in the DBMS?

There are two major integrity rules that exist in the DBMS.

Entity Integrity: This states a very important rule that the value of a Primary key can never have a NULL value.

Referential Integrity: This rule is related to the Foreign key which states that either the value of a Foreign key is a NULL value or it should be the primary key of any other relation.

35. What is E-R model in the DBMS?

E-R model is known as an Entity-Relationship model in the DBMS which is based on the concept of the Entities and the relationship that exists among these entities.

36. What is a functional dependency in the DBMS?

This is basically a constraint that is useful in describing the relationship among the different attributes in a relation.

Example: If there is some relation ‘R1’ which has 2 attributes as Y and Z then the functional dependency among these 2 attributes can be shown as Y->Z which states that Z is functionally dependent on Y.

37. What is 1NF in the DBMS?

1NF is known as the First Normal Form.

This is the easiest form of the normalization process which states that the domain of an attribute should have only atomic values. The objective of this is to remove the duplicate columns that are present in the table.

38. What is 2NF in the DBMS?

2NF is the Second Normal Form.

Any table is said to have in the 2NF if it satisfies the following conditions:

A table is in the 1NF.

Each non-prime attribute of a table is said to be functionally dependent in totality on the primary key.

39. What is 3NF in the DBMS?

3NF is the Third Normal Form.

Any table is said to have in the 3NF if it satisfies the following 2 conditions:

A table is in the 2NF.

Each non-prime attribute of a table is said to be non-transitively dependent on every key of the table.

40. What is BCNF in the DBMS?

BCNF is the Boyce Codd Normal Form which is stricter than the 3NF.

Any table is said to have in the BCNF if it satisfies the following 2 conditions:

A table is in the 3NF.

For each of the functional dependencies X->Y that exists, X is the super key of a table.

41. What is a CLAUSE in terms of SQL?

This is used with the SQL queries to fetch specific data as per the requirements on the basis of the conditions that are put in the SQL. This is very helpful in picking the selective records from the complete set of records.

For Example, There is a query that has a WHERE condition or the query with the HAVING clause.

42.How can you get the alternate records from the table in the SQL?
If you want to fetch the odd numbers then the following query can be used:

```
SELECT EmpId from (SELECT rowno,EmpId from Emp)
WHERE mod(rowno,2)=1;
```

If you want to fetch the even numbers, then the following query can be used:

```
SELECT EmpId from (SELECT rowno,EmpId from Emp)
WHERE mod(rowno,2)=0;
```

43. How is the pattern matching done in the SQL?

Answer: With the help of the LIKE operator, pattern matching is possible in the SQL.'%' is used with the LIKE operator when it matches with the 0 or more characters, and '_' is used to match the one particular character.

Example:

```
SELECT * from Emp WHERE name like 'b
```

```
%';
```

```
SELECT * from Emp
WHERE name like 'hans_';
```

44. What is a join in the SQL?

A Join is one of the SQL statements which is used to join the data or the rows from 2 or more tables on the basis of a common field/column among them.

45. What are the different types of joins in SQL?

There are 4 types of SQL Joins:

Inner Join: This type of join is used to fetch the data among the tables which are common in both tables.

Left Join: This returns all the rows from the table which is on the left side of the join but only the matching rows from the table which is on the right side of the join.

Right Join: This returns all the rows from the table which is on the right side of the join but only the matching rows from the table which is on the left side of the join.

Full Join: This returns the rows from all the tables on which the join condition has been put and the rows which do not match hold null values.

46. Explain the Stored Procedure.

A Stored Procedure is a group of SQL statements in the form of a function that has some unique name and is stored in relational database management systems(RDBMS) and can be accessed whenever required.

47. What is RDBMS?

RDBMS is the Relational Database Management System which contains data in the form of the tables and data is accessed on the basis of the common fields among the tables.

48. What are the different types of relationships in the DBMS?

A Relationship in DBMS depicts an association between the tables.

Different types of relationships are:

One-to-One: This basically states that there should be a one-to-one relationship between the tables i.e. there should be one record in both the tables.
One-to-Many: This states that there can be many relationships for one i.e. a primary key table hold only one record which can have many, one, or none records in the related table.
Many-to-Many: This states that both the tables can be related to many other tables.
49. What do you mean by Entity type extension?

Compilation of similar entity types into one particular type which is grouped together as an entity set is known as entity type extension.

50. What is conceptual design in dbms?

Conceptual design is the first stage in the database design process. The goal at this stage is to design a database that is independent of database software and physical details. The output of this process is a conceptual data model that describes the main data entities, attributes, relationships, and constraints of a given problem domain.